Software Tools and Technologies Laboratory
Latex Lab Activity - 3: Beamer

Anand M. Baswade[1] Arindam Chakraborty[2]

[1]Faculty of Computer Science and Engineering
Indian Institute of Technology, Bhilai

[2]M.Tech Student of Computer Science and Engineering
Indian Institute of Technology, Bhilai

September 16, 2025



Table of Contents

1. Introduction
2. Why Beamer?
3. Features
4. Highlighting Texts
5. Mathematical Concepts
6. Linear Search Algorithm
7. Images
8. Important Equations
9. Conclusion
10. References
11. Thank You



Table of Contents

1. Introduction
2. Why Beamer?
3. Features
4. Highlighting Texts
5. Mathematical Concepts
6. Linear Search Algorithm
7. Images
8. Important Equations
9. Conclusion
10. References
11. Thank You



Introduction to LaTeX Beamer

- LaTeX Beamer is a powerful class for creating presentations.
- It allows:
  - Professional-quality slides
  - Mathematical equations
  - Figures, tables, and references
  - Animations and overlays
- Advantages:
  - Consistent formatting
  - Easy handling of citations and references
  - Cross-platform PDF output



Table of Contents

1. Introduction
2. Why Beamer?
3. Features
4. Highlighting Texts
5. Mathematical Concepts
6. Linear Search Algorithm
7. Images
8. Important Equations
9. Conclusion
10. References
11. Thank You



Why Use Beamer?

- Beamer is designed for creating professional presentations directly in LaTeX.



Why Use Beamer?

- Beamer is designed for creating professional presentations directly in LaTeX.
- Consistent and elegant design



Why Use Beamer?

- Beamer is designed for creating professional presentations directly in LaTeX.
- Consistent and elegant design
- Full support for mathematical equations



Why Use Beamer?

- Beamer is designed for creating professional presentations directly in LaTeX.
- Consistent and elegant design
- Full support for mathematical equations
- Easy management of references and citations



Why Use Beamer?

- Beamer is designed for creating professional presentations directly in LaTeX.
- Consistent and elegant design
- Full support for mathematical equations
- Easy management of references and citations
- Portable PDF output (works everywhere)



Why Use Beamer?

- Beamer is designed for creating professional presentations directly in LaTeX.
- Consistent and elegant design
- Full support for mathematical equations
- Easy management of references and citations
- Portable PDF output (works everywhere)
- Highly customizable (themes, colors, backgrounds, animations)



Why Use Beamer?

- Beamer is designed for creating professional presentations directly in LaTeX.
- Consistent and elegant design
- Full support for mathematical equations
- Easy management of references and citations
- Portable PDF output (works everywhere)
- Highly customizable (themes, colors, backgrounds, animations)
- Free and open-source — no licensing issues



Comparison with Other Tools

Feature	Beamer	PowerPoint
Equations		
Citations		
Portability		
Cost		
Customization		



Comparison with Other Tools

Feature	Beamer	PowerPoint
Equations	Excellent	Limited
Citations		
Portability		
Cost		
Customization		



Comparison with Other Tools

Feature	Beamer	PowerPoint
Equations	Excellent	Limited
Citations	Automatic	Manual
Portability		
Cost		
Customization		



Comparison with Other Tools

Feature	Beamer	PowerPoint
Equations	Excellent	Limited
Citations	Automatic	Manual
Portability	PDF (stable)	File-dependent
Cost		
Customization		



Comparison with Other Tools

Feature	Beamer	PowerPoint
Equations	Excellent	Limited
Citations	Automatic	Manual
Portability	PDF (stable)	File-dependent
Cost	Free	Proprietary
Customization		



Comparison with Other Tools

Feature	Beamer	PowerPoint
Equations	Excellent	Limited
Citations	Automatic	Manual
Portability	PDF (stable)	File-dependent
Cost	Free	Proprietary
Customization	High (code-based)	Moderate (GUI-based)



Table of Contents

1. Introduction
2. Why Beamer?
3. Features
4. Highlighting Texts
5. Mathematical Concepts
6. Linear Search Algorithm
7. Images
8. Important Equations
9. Conclusion
10. References
11. Thank You



Key Features of LaTeX Beamer

Presentation Design

- Predefined professional themes for elegant slides

Content Capabilities

References & Automation

Key Features of LaTeX Beamer

Presentation Design

- Predefined professional themes for elegant slides
- Customizable colors, fonts, and backgrounds

Content Capabilities

References & Automation

Key Features of LaTeX Beamer

Presentation Design

- Predefined professional themes for elegant slides
- Customizable colors, fonts, and backgrounds
- High-quality PDF output across platforms

Content Capabilities

References & Automation

Key Features of LaTeX Beamer

Presentation Design

- Predefined professional themes for elegant slides
- Customizable colors, fonts, and backgrounds
- High-quality PDF output across platforms

Content Capabilities

- Full support for complex mathematical equations

References & Automation

Key Features of LaTeX Beamer

Presentation Design

- Predefined professional themes for elegant slides
- Customizable colors, fonts, and backgrounds
- High-quality PDF output across platforms

Content Capabilities

- Full support for complex mathematical equations
- Easy insertion of tables, figures, and graphics

References & Automation

Key Features of LaTeX Beamer

Presentation Design

- Predefined professional themes for elegant slides
- Customizable colors, fonts, and backgrounds
- High-quality PDF output across platforms

Content Capabilities

- Full support for complex mathematical equations
- Easy insertion of tables, figures, and graphics
- Step-by-step content reveal using overlays

References & Automation

Key Features of LaTeX Beamer

Presentation Design

- Predefined professional themes for elegant slides
- Customizable colors, fonts, and backgrounds
- High-quality PDF output across platforms

Content Capabilities

- Full support for complex mathematical equations
- Easy insertion of tables, figures, and graphics
- Step-by-step content reveal using overlays

References & Automation

- Integration with BibTeX for automatic citations

Key Features of LaTeX Beamer

Presentation Design

- Predefined professional themes for elegant slides
- Customizable colors, fonts, and backgrounds
- High-quality PDF output across platforms

Content Capabilities

- Full support for complex mathematical equations
- Easy insertion of tables, figures, and graphics
- Step-by-step content reveal using overlays

References & Automation

- Integration with BibTeX for automatic citations
- Efficient workflow for large, structured presentations

Table of Contents

1. Introduction
2. Why Beamer?
3. Features
4. Highlighting Texts
5. Mathematical Concepts
6. Linear Search Algorithm
7. Images
8. Important Equations
9. Conclusion
10. References
11. Thank You



Blocks and Highlights Example

Some important text can be highlighted to attract attention.

Remark

This is a normal block. It wraps the text in a box with the same style as the presentation. The text after the block title is the content.

Important Theorem

This is an alert block. It contrasts with the regular presentation style, useful to emphasize key results or theorems.

Examples

This is an example block. It is similar to a normal block, but has a different style (less contrasting than alertblock). Perfect for examples.



Table of Contents

1. Introduction
2. Why Beamer?
3. Features
4. Highlighting Texts
5. Mathematical Concepts
6. Linear Search Algorithm
7. Images
8. Important Equations
9. Conclusion
10. References
11. Thank You



Definition

Definition

A prime number is a natural number greater than 1 that has no positive divisors other than 1 and itself.



Theorem & Lemma

Theorem

Fundamental Theorem of Arithmetic: Every integer greater than 1 is either a prime or can be written as a unique product of prime numbers.

Lemma

If a prime p divides a product ab, then p divides a or p divides b.



Table of Contents

1 Introduction
2 Why Beamer?
3 Features
4 Highlighting Texts
5 Mathematical Concepts
6 Linear Search Algorithm
7 Images
8 Important Equations
9 Conclusion
10 References
11 Thank You



Linear Search Algorithm

Input: Array A of size n, key to search
Output: Index of key in A, or -1 if not found

```
for i ← 0 to n − 1 do
    if A[i] = key then
        return i;
    end if
end for
return -1;
```

Algorithm 1: Linear Search



Corollary & Proof

Corollary

The factorization of any integer into primes is unique up to the order of the factors.

Proof

Suppose n has two different prime factorizations. By the lemma, every prime in one factorization divides one of the primes in the other factorization. This leads to a contradiction, proving uniqueness.



Table of Contents

1. Introduction
2. Why Beamer?
3. Features
4. Highlighting Texts
5. Mathematical Concepts
6. Linear Search Algorithm
7. Images
8. Important Equations
9. Conclusion
10. References
11. Thank You



Including an Image in a Block

Example Image



Figure 1: This is a sample image included in a Beamer slide.

Table of Contents

1 Introduction
2 Why Beamer?
3 Features
4 Highlighting Texts
5 Mathematical Concepts
6 Linear Search Algorithm
7 Images
8 Important Equations
9 Conclusion
10 References
11 Thank You



Key Equations in Signal Processing

Equation 1: Fourier Transform

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt$$

Equation 2: Convolution Integral

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$$

Equation 3: Transfer Function of LTI System

$$H(f) = \frac{Y(f)}{X(f)} = \int_{-\infty}^{\infty} h(t) e^{-j2\pi ft} dt$$



Table of Contents

1 Introduction
2 Why Beamer?
3 Features
4 Highlighting Texts
5 Mathematical Concepts
6 Linear Search Algorithm
7 Images
8 Important Equations
9 Conclusion
10 References
11 Thank You



Conclusion

- LaTeX Beamer is a powerful tool for creating professional presentations.



Conclusion

- LaTeX Beamer is a powerful tool for creating professional presentations.
- It supports complex mathematical equations, figures, tables, and references.



Conclusion

- LaTeX Beamer is a powerful tool for creating professional presentations.
- It supports complex mathematical equations, figures, tables, and references.
- Overlays and incremental reveals help guide the audience's focus.



Conclusion

- LaTeX Beamer is a powerful tool for creating professional presentations.
- It supports complex mathematical equations, figures, tables, and references.
- Overlays and incremental reveals help guide the audience's focus.
- Highly customizable themes, colors, and backgrounds make slides visually appealing.



Conclusion

- LaTeX Beamer is a powerful tool for creating professional presentations.
- It supports complex mathematical equations, figures, tables, and references.
- Overlays and incremental reveals help guide the audience's focus.
- Highly customizable themes, colors, and backgrounds make slides visually appealing.
- Ideal for academic and technical presentations due to its precision and portability.



Table of Contents

1 Introduction

2 Why Beamer?

3 Features

4 Highlighting Texts

5 Mathematical Concepts

6 Linear Search Algorithm

7 Images

8 Important Equations

9 Conclusion

10 References

11 Thank You



References I

This presentation is based on the following references:

- [1] for basic LaTeX concepts.
- [2] for fundamental TeX understanding.
- [4] for advanced LaTeXusage.
- [5]for best practices in document preparation.
- [3] for graphics and presentation guidelines in LaTeX.


Michel Goossens, Frank Mittelbach, and Alexander Samarin.
The latex companion.
In *Proceedings of the TeX Users Group Meeting*. TUG, 1994.


Donald E. Knuth.
The texbook.
Addison-Wesley, 1984.



References II


Helmut Kopka and Patrick W. Daly.
A Guide to LaTeX.
Addison-Wesley, 4th edition, 2003.


Leslie Lamport.
LaTeX: A Document Preparation System.
Addison-Wesley, 2nd edition, 1994.


Frank Mittelbach and Michel Goossens.
The LaTeX Companion.
Addison-Wesley, 2nd edition, 2004.



Table of Contents

1 Introduction

2 Why Beamer?

3 Features

4 Highlighting Texts

5 Mathematical Concepts

6 Linear Search Algorithm

7 Images

8 Important Equations

9 Conclusion

10 References

11 Thank You



Thank You!

Any Questions?



